

to improve operations. But due to the B2B intermediaries, customers need to pay a high price. So, the industry should also consider use of blockchain.

The tourism sector has started using blockchain to change the way their data is stored, used, and secured. The sector expects blockchain to reshape the industry through transparent and secured transactions without the involvement of any third party between the hosts and guests, digital travel ledgers, faster identity management of customers, and much more. A report published by SITA (Société Internationale de Télécommunications Aéronautiques) declares that by 2021 year's end, 59 percent of airlines would have initiated implementation of blockchain in their working.

Blockchain and cryptocurrency

Blockchain is a distributed, decentralised data structure that serves as a proof of work for the occurrence of events in a specific order. This digital ledger is immutable and stores the encrypted data. For example, when a transaction is made online, that event is represented as a block and is broadcast to peer-to-peer (P2P) network (computers/nodes).

The mechanism employs an encryption method known as cryptography, which uses known algorithms to validate the transaction and user's status. The verified transaction can involve cryptocurrency, contracts, records, or other information. The transaction is combined with other transactions for verification. Then a new block of data is created for the ledger. The new block is added to the existing, permanent, and unalterable blockchain, and there is no way of erasing a data. The transaction is now finished without the involvement of any bank. Each transaction in the form of a block is time stamped, to pin the event to the real time, and many such blocks are linked together forming a chain, hence the name blockchain. These chains cryptographically link the past events to the current events, thus preserving their order under the latest time stamp.

Cryptocurrency does not require an intermediary like a bank to trans-



Fig. 1: AI increases global GDP (Image credit: pwc.com)

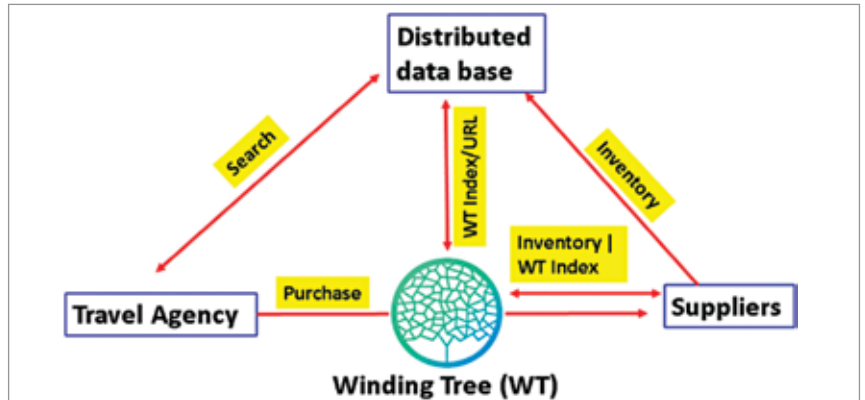


Fig. 2: Winding Tree decentralised ecosystem

act money. Instead, it makes use of blockchain technology. It has been in news, unfortunately, for enabling illegal purchases, money laundering, and tax evasion, which has created a wrong perception in the society. The legitimacy of dealing with this digital currency was difficult as there were no hard and fast rules to unveil the credentials of the involved criminals. However, after following the rules and guidelines set by various governments and the international Financial Action Task Force (FATF), cryptocurrency exchanges seem to be now able to keep the phenomenon under control.

Tourism industry

Planning travel often makes us flummoxed due to the complexity and the time involved in buying air tickets, managing transfers, booking rentals, hotels, and many more involved activities. Use of blockchain can bring safety and transparency to a plethora of organisations that are involved in a travel plan. A traveller is required to show passport or a valid ID at multiple stages of his journey such as booking, boarding, and check-in at the hotel. Blockchain could generate a single travel token for each traveller and this information would be universally accessible to multiple airports, thus

making the ID verification process simple, secure, and efficient.

Making finance settlements between travel agents, hotels, and other travel-related firms is an extremely complicated process, especially when it comes to overseas transactions that involve currency exchange. Blockchain can streamline this process by reducing the intermediaries that can intervene or delay the settlement timing and increasing the overall transparency and the trust to a whole new level.

Baggage tracking is another challenge that requires keeping track of the travellers' luggage as it changes hands from airlines to the ground handling services at the airports. The luggage data can be stored and processed by blockchain that allows passengers to track their luggage over the cloud and thus reduce baggage mishandling. Blockchain will allow all parties to exchange information even if they are not related to the same airline or same subdivision, thus improving the accuracy of baggage tracking across the industry.

Blockchain in use

Winding Tree is a decentralised open source business-to-business (B2B) travel marketplace. It is a platform